



Full Name: Zamin Shaheed Siddiqui

Designation

Director MAHQ Biological Research Centre, University of Karachi

Professor since April 2023 till now at the University of Karachi

Associate Editor in “Frontier in Plant Sciences” Abiotic stress Section

Chief Editor International Journal of Biology Research

Institution

Stress Physiology and Phenomic Center, Department of Botany, University of Karachi,

Karachi-75270 Pakistan. Email: zaminss@uok.edu.pk, zaminsiddiqui@gmail.com

Contact: 03073607210, +9221-3483726,

International Information

Korean Researcher Number: 1115-5945, Canadian Immigration # T523023972

Links:

1. [Zamin Shaheed Siddiqui - Google Scholar](#)
2. <https://www.researchgate.net/profile/Zamin-Shaheed-Siddiqui>
3. [Loop | Zamin Shaheed Siddiqui \(frontiersin.org\)](#),

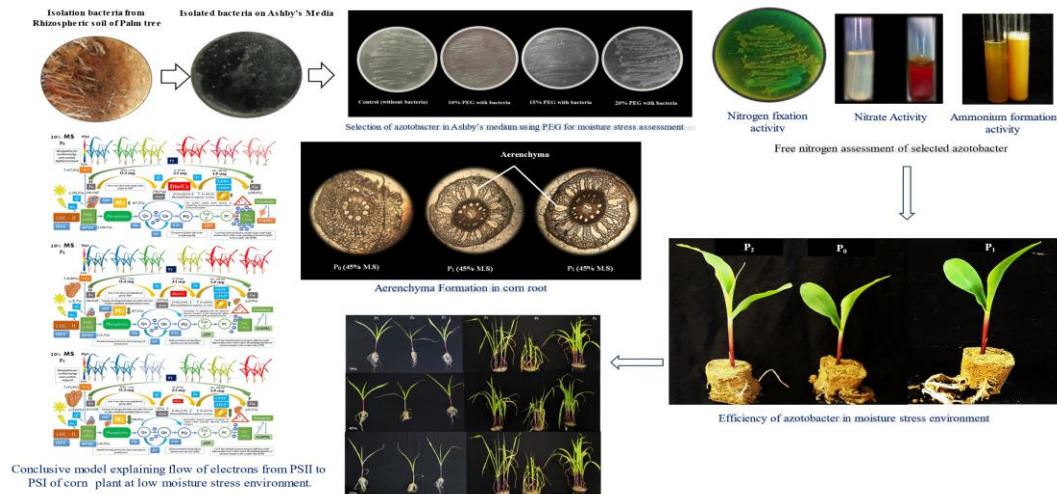
Total Citation: 1275, h- index: 23, i -10 Index: 39

RESEARCH PROJECT, INTEREST AND ACHIVEMENTS

Plant Phenomics has become a very important field of research. In most of the advanced countries, they have already established phenomics facility. I established a phenomics center at the Department of Botany, University of Karachi (Fig attached). This work is supported by a **international grant** from the **NEXT- GENERATION BIO-GREEN 21 Program No. PJ009008 and PJ009505, Rural Development Administration, Suwon, Republic of Korea Completed 2013-16**
(Nearly US \$ 1, 20, 000/= US Dollars)

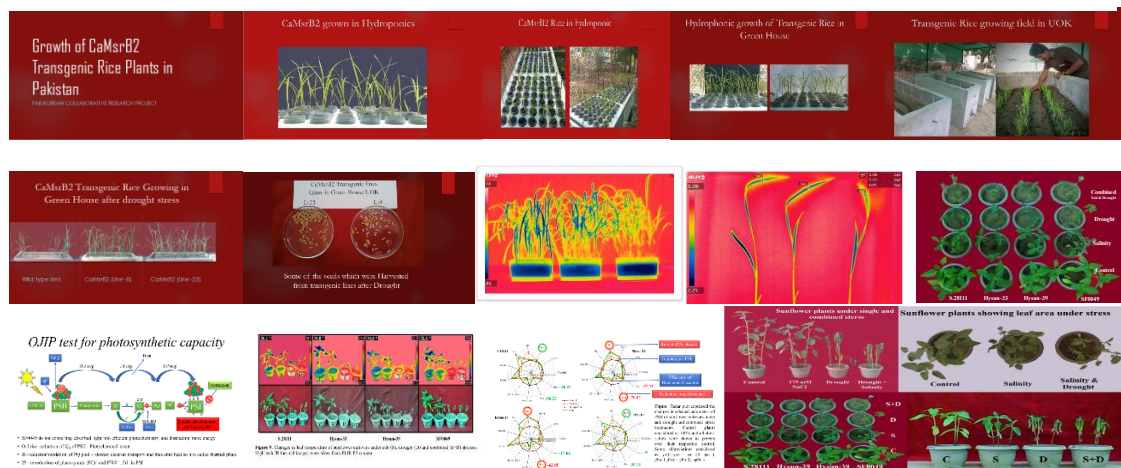
At national level I got two projects, one is completed in 2019,

- 1) Project entitled “Phenotyping of some oil seed crops using IR thermography. **National Research Program for Universities, HEC, Pakistan. HEC PROJECT ID # 6586 dated 2016-17”**
 - 2) Project NRPU # **Ref No. 20-17449/NRPU/R&D/HEC/2021 2021-2024 is now operational and running.** Title of the project “Sustainable use of natural system (salt tolerant endophytes) to combat desertification and salinization and their evaluation by non-destructive phenotypic methods.”
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The Photochemical model displays fluxes in various shapes (Komal et al., 2024)

International collaborative work during the year 2016-2023



Professional Training

Post Doc Fellowship 2012-2013: National Center for GM Crop, National Academy of Agricultural Sciences, Rural Development Administration Republic of Korea

Post Doc Collaborative work 2013-2016. Physiological performance of Korean rice in Pakistan

Ph.D. June 2010: Biochemical Responses of *Halopyrum mucronatum* L. Stapf. seed during germination”. **Dr. M. Ajmal Khan**, Professor & Director (Institute of Sustainable Halophyte Utilization University of Karachi).

CONFERENCES AND SEMINARS

- Chair Session “*International Conference on Saline Resources Management in Context of Global Climate Changes*” Feb 23-25, 2023
- One Day Seminar on “*Biodiversity: Conservation, Issues & Challenges*” 22ND Feb 2023 (Organizer)
- Seminar on Ecosystem Services of Mangroves 4th August 2022
- Oral Presentation 1st International Conference on Plant Protection Sciences (ICPPS-2022) Sindh Agriculture University TandoJam
- One Day Seminar on Plant and Environmental Protection 2nd Feb 2022.
- 6th International and 16th National Conferences “*Pakistan Botanical Society*” March 2018
- **Invited Speaker** at 3-Day workshop on “*Advancement Of Agricultural Biotechnology And Regulatory Affairs*” at Forman Christian College A Chartered University (25-27September 2017)
- **Focal person for plant physiology in** “14th national and 5th International conference of Botany” *Climate change and phyto-diversity: opportunities and challenges* Held on 2016.
- Shameel Memorial Seminar on potential Seaweed Market in Pakistan May 13th & ne 14th 2015
- International conference on salinity 1996 held on 1996 at Department of Botany University of Karachi.
- National Conference of Plant Scientist 2002
- 9th National Conference of Plant Scientist 2006
- **Resource person of HEC workshop entitled** “*In-Service Training course for College Teacher* 9-16th Dec, 2004-05.
- HEC workshop entitled “*Current advancement in Plant Science*” at Karachi University. 2004-05
- Poster Presentation on International Conference on Environmental Horizons May 2005-2006 held at Department of Chemistry, University of Karachi
- Recent Trend in Plant Disease Management 20th Dec, 2005-06
- Participant in Two-Day National Conference on Life Sciences (13-14 Nov 2006).
- 2007 Participated various seminars in South Korea (Sound Induced Physiological Changes in rice genotypes).

ACADEMIC AWARDS AND HONORS

1994: Merit Certificate from Botkarian, Department of Botany, University of Karachi.
1997: Sindh Public Service Commission. (A competitive Examination held on December, 1996 by Government of Sindh, Pakistan)
2002: Nishan-e-Thaniat from Karachi University Teacher Association (KUTS)
2004-05: **Resource person** HEC workshop at regional office Karachi
2005: Cash Awards for Research Incentive during 2004-onward
2006: Selected Participant for eight meeting of National Core of Life Sciences (HEC)
2006-07: International Research Support Initiative Program Scholarship funded by

Higher Education Commission. Pakistan

2007-2008: Visiting Scholar from HEC visited to RDA, NIAB Suwon, South Korea

2009-till date: Member of National Academy of Young Scientist Pakistan

2010: Head Examiner in Practical IV Paper IV (Karachi Region Colleges)

2010-2011: RPA, Research Productivity Award by Pakistan Council for Science and Technology

2011-2012: RPA, Research Productivity Award by Pakistan Council for Science and Technology

2012-2013: International Post Doc Fellowship (NCGC South Korea)

2013: Head Examiner in Theory Exam Paper IV (Karachi Region Colleges)

2016: Head Examiner in Theory Exam Paper IV (Karachi Region Colleges)

2023: Selected as top Professor of the merit list

Administrative Responsibilities

- 1) Campus Security Advisor 2017-18
- 2) Convenor Transport Committee 2017-18
- 3) Member of Land Utilization Committee KU 2017-18
- 4) Member Board of Studies 2011-till Now
- 5) Member Board of Faculty Science 2011-2016, 2021 till now
- 6) Member of Senate University of Karachi April 2023 till now
- 7) Member of Academic Council University of Karachi April 2023 till now

SELECTED PUBLICATIONS

1. Zamin Shaheed Siddiqui, Komal Nida, Jung-Il Cho, Yusra Rehman, Zainul Abideen.2024. Physiological and photochemical profiling of soybean plant using biological and chemical methods of treatment against biotic stress management. *Plant Physiology and Biochemistry* [10.1016/j.plaphy.2024.108454](https://doi.org/10.1016/j.plaphy.2024.108454)
2. Komal Nida, Zamin Shaheed Siddiqui, Zeeshan Ahmed Salman, Asma Aftab, Rubina Abid, Zainul Abideen, Manzer H Siddiqui. 2024. Scrutinize the integrated role of *Azotobacter vinelandii* in nitrogen assimilation, photosystem II functionality and aerenchyma formation of *Zea mays* under moisture stress environment. *Plant Stress*. DOI.ORG/10.1016/STRESS.2024.100378
3. Asma Hanif, Zainul Abideen, Khwaja Ali Hasan, Basit Ali, Ali El-Keblawy, Muhammad Umar, Yamna Rao, Zamin Shaheed Siddiqui. 2023. Application of microbial antagonists as alternatives to chemical pesticides: A novel approach for enhancing plant growth and resilience of agricultural land. *Land Degradation Development*. DOI: [10.1002/ldr.4859](https://doi.org/10.1002/ldr.4859).
4. Khadija Irshad, Zamin Shaheed Siddiqui, Jianjun Chen, Yamna Rao, Hafiza Hamna Ansari, Danish Wajid, Komal Nida and Xiangying Wei. 2023. Bio-priming with salt tolerant endophytes improved crop tolerance to salt stress *via* modulating photosystem II and antioxidant activities in a sub-optimal environment. *Frontier in Plant Sciences*, doi.org/10.3389/fpls.2023.1082480.
5. Fiza Ali, Zamin Shaheed Siddiqui, Hafiza Hamna Ansari, Urooj Zafar, Danish Wajid, Muhammad Waseem Abbasi, Yamna Rao.2023. Halophilic soil microbial strains improve the moisture stress tolerance in oilseed crop by sustaining Photosystem II functionality. *Plant Physiology and Biochemistry* 196: 10–22.

6. Mammad Azeem, Robina Sultana, Athar Mahmood, Muhammad Qasim, Zamin Shaheed Siddiqui, Sahar Mumtaz, Talha Javed, Muhammad Umar, M. Yousuf Adnan, Manzer H. Siddiqui. 2023. Ascorbic and Salicylic Acids Vitalized Growth, Biochemical Responses, Antioxidant Enzymes, Photosynthetic Efficiency, and Ionic Regulation to Alleviate Salinity Stress in Sorghum bicolor. *Journal of Plant Growth Regulation* <https://doi.org/10.1007/s00344-023-10907-2>
7. Zamin Shaheed Siddiqui¹, Naeema Asim and Zaheer Uddin. 2022. Effect of combined organic and halo priming on physiological tolerance in Flax plants under saline environment and their evaluation by newly developed predicted regression model. *Russian Journal of Plant Physiology* Vol. 69:102.
8. Kainat Amin Bora, Saud Hashmi, Faisal Zulfiqar, Zainul Abideen, Haibat Ali, Zamin Shaheed Siddiqui and Kadambot H. M. Siddique 2022. Recent progress in bio-mediated synthesis and applications of engineered nanomaterials for sustainable agriculture *Frontier in Plant Sciences*, doi.org/10.3389/fpls.2022.99950.
9. Fiza Ali, Zamin Shaheed Siddiqui, Yamna Rao, Hafiza Hamna Ansari, Urooj Zafar, Danish Wajid, and Muhammad Waseem Abbasi ²⁰²². Halophilic soil microbial strains sustain Photosystem II functionality and thus improve physiological tolerance in oilseed cultivars under moisture stress environment. *Frontier in Plant Sciences* doi: 10.3389/fpls.2022.913825.
10. Fiza A, Xiangying W, Siddiqui ZS, Jianjun C, Ansari HH, Danish W, Shams ZI, Abbasi MW, Urooj Z (2022). Scrutinizes the Sustainable Role of Halophilic Microbial Strains on Oxygen-Evolving Complex, Specific Energy Fluxes, Energy Flow and Nitrogen Assimilation of Sunflower Cultivars in a Suboptimal Environment *Frontier in Plant Sciences* doi.10.3389.2022.913825.
11. Basit Ali, Muhammad Umar, Muhammad Azeem, Zaheer Uddin and Zamin Shaheed Siddiqui 2022. A Comparative physiological evaluation of halophytic grass (*Cenchrus penisetiformis*) vs newly developed wheat variety (AZRC-DK-84) in abiotic stress environment. *Acta Physiol. Plant.* 44:81 doi.10.1007/s11738-022-03421-7
12. Maria Hasnain¹, Neelma Munir¹, Zamin Shaheed Siddiqui², Faraz Ali³, Ali El-Keblawy⁴ and Zainul Abideen 2022. Integral approach for the evaluation of sugar cane bio-waste molasses and effects on algal lipids and biodiesel production. *Waste and Biomass Valorization* doi:10.1007/s.12649-022-01684-0
13. Siddiqui ZS, Gang-Seob Lee, Woosuk Cho, Mi-Jeong Jeong, Soo-Chul Park, Taek-Ryoun Kwon, Faisal Zulfiqar, Muhammad Umar, Zain ul Abedin, Zaheer Uddin, Hafiza Hamna Ansari, Danish Wajid, Jung-Il Cho (2022). Physiological aspects of germination and early seedling establishment of *Pleurotus sajor-caju* glyceraldehyde-3-phosphate dehydrogenase expressing transgenic rice in saline environment. *Frontier in Plant Sciences* doi:10.3389/fpls.2021.7678264.
14. Ansari HH, Siddiqui A, Wajid D, Tabassum S, Umar M, Siddiqui ZS (2022). Profiling of energy compartmentalization in photosystem II (PSII), light harvesting complexes and specific energy fluxes of primed maize cultivar (P1429) under salt stress environment. *Plant Physiology and Biochemistry* 170:296-306.
15. Siddiqui ZS, Xiangying Wei, Muhammad Umar, Zainul Abideen, Faisal Zulfiqar, Jianjun Chen, Asma Hanif, Shahnaz Dawar, Daniel Anthony Dias and Roomana Yasmeen¹

Scrutinizing the application of saline endophyte to enhance salt tolerance in rice and maize plants. *Frontier in Plant Sciences* doi:10.3389/fpls.2021.770084

16. Siddiqui ZS, Amrina Habib, M Umar, Z.I Shams.2022 Effect of *Phythium aphanidermatum* (root rot pathogen) on the physiology of *Luffa cylindrica* (Sponge gourd) and its management by microbial antagonists. *South African Journal of Botany* 146:614-613
17. Siddiqui ZS, Sung-Dug Oh, Eun-Ji Kim, Ye-Jin Jang, Soo-Chul Park, Jung-Il Cho, Taek-Ryoun Kwon, Danish Wajid, Hafiza Hamna Ansari (2021). Physiological and photochemical performance of CaMsrb2 expressing transgenic rice in saline habitat. *Plant Physiology and Biochemistry* 167:198-209.
18. Siddiqui ZS., Fizza Ali and Z.Uddin (2021). Sustainable effect of a symbiotic nitrogen-fixing bacterium *Sinorhizobium meliloti* on nodulation and photosynthetic traits of four leguminous plants under low moisture stress environment. *Letter in Applied Microbiology* Feb 16. doi: 10.1111/lam.13463.
19. Fizza A, M. Umar and Siddiqui Z.S.2021. Comparative physiological assessment of some edible oil seed crops under drought stress environment using fluorescence and IR imaging techniques. *Pak. J. Bot.* 53:4
20. Faisal Zulfiqar, Jianjun Chen, Patrick M. Finnegan, Muhammad Nafees, Adnan Younis, Narmeen Shaukat , Nadeem Latif , Zainul Abideen, Abbu Zaid, Ali Raza, Zamin Shaheed Siddiqui and Karim Ben Hamed (2021). Foliar Application of Trehalose or 5-Aminolevulinic Acid Improves Photosynthesis and Biomass Production in Drought Stressed *Alpinia zerumbet*. *Agriculture* 2021, 11, 908.
21. Dawar S., M. Tariq, ZS. Siddiqui and N. Bashir 2020. Impact of Chemical Priming on Root Infecting Fungi, Photosynthesis and Yield Components of Cowpea and Mung Bean. *Pak.J.Bot.* 52: 6
22. Kanwa S, Marium T, S. Dawar and ZS Siddiqui (2020) Electromagnetic radiations and microbial antagonists' potential on yield and physiology of sunflower *Pak. J Bot.* DOI:10.30848/PJB2022-2(9)
23. Umar M. and Siddiqui Z.S. 2020. Florescence Assessment of Sunflower Genotypes against drought stress environment. *Pak. J. Bot.* 52: 6
24. Siddiqui ZS, KH Lee, Y-S Kim, G-S Lee, J I Cho, S-C Park. 2019. Biochemical Changes of CaMsrb2 Expressing Transgenic Rice Seed during Germination in Heavy Metal Stress Environment. *Plant Breed. Biotech.* 7:287~294
25. Umar M. Uddin Zaheer and Siddiqui Z.S. 2019: Responses of photosynthetic apparatus in sunflower cultivars to combined drought and salt stress PSII efficiency and antioxidant activities of some sunflower genotypes in combined salt and drought stress. *Photosynthetica* 57: 627-639
26. Roomana Y, Siddiqui Z.S. 2018: Ameliorative effects of *Trichoderma harzianum* on monocot crops under hydroponic saline environment. *Acta Physiologiae Plantarum* 40:4 doi.10.1007/s11738-017-2579-2
27. Roomana Y, Siddiqui Z.S. 2018: Physiological responses of crop plants against *Trichoderma harzianum* in saline environment. *Acta Bot. Croat.* 76: 154-162.
28. Umar M. and Siddiqui Z.S. 2018: Physiological responses of sunflower genotypes under combined stress environment. *Acta. Bot. Croat.* 77: 36–44.
29. Umar M. and Siddiqui Z.S. 2017: Comparison of two most commonly used screening techniques for salt tolerance in crop plants. *Pak. J. Bot* 49: 2103-2113

30. Amrina H, Shahzad S, Siddiqui Z.S. 2017: Photochemistry of *Luffa cylindrica* (L.) Roem under fungal biocontrol interaction. *Photosynthetica*. 56: 743-749.
31. Siddiqui ZS, Shahid H, Cho J-Il, Park S-H, Ryu T-H, Park S-C. 2016: Physiological responses of two halophytic grass species under drought stress environment *Acta Bot. Croat.* 75: 31–38
32. Kwon T-R Kim, K-H, Yoon H-J, Lee S-K, Kim BK, Siddiqui ZS. 2015: Phenotyping of Plants for Drought and Salt Tolerance Using Infra-Red Thermography. *Plant Breed. Biotech.* 3: 299-307.
33. Siddiqui ZS, Cho J-Il, Park D-B, Lee G-S, Ryu T-H, Shahid H, Umar M, Park S-C. 2015: Field assessment of CaMsrB2 transgenic lines in a drought stress environment. *Turk. J. Bot.* 39: 973-981
34. Cho JI, HM Lim, ZS Siddiqui, SH Park, AR Kim, TR Kwon, SK Lee, SC Park, MJ Jeong, GS Lee. 2014: Over-expression of *PsGPD*, a mushroom glyceraldehyde-3-phosphate dehydrogenase gene, enhances salt tolerance in rice plants. *Biotechnology Letters*. 36: 1641-1648.
35. Siddiqui ZS, JI Cho, TR Kwon, BO Ahn, KS Lee, MJ Jeong, TH Ryu, SK Lee, SC Park, SH Park. 2014: Physiological mechanism of drought tolerance in transgenic rice plants expressing *Capsicum annuum* methionine sulfoxide reductase B2 (*CaMsrB2*) gene. *Acta Physiologiae Plantarum*. 36: 1143-1153.
36. Siddiqui ZS, JI Cho, SH Park, TR Kwon, BO Ahn, GS Lee, MJ Jeong, KW Kim, SK Lee, and SC Park. 2014: Phenotyping of rice in salt stress environment using high-throughput infrared imaging. *Acta Bot. Croat.* 73: 149–158.
37. Shahid H, JI Cho, SC Park, TH Ryu, SH Park and Zamin Shaheed Siddiqui. 2014: Physiological performance of two Korean rice cultivars under saline environment at seedling stage. *Int. J. Biol. Biotech.* 11: 57-65.
38. Jeong MJ, JI Cho, SH Park, KH Kim, SK Lee, TR Kwon, SC Park, and ZS Siddiqui. 2014: Sound frequency induces drought tolerance in rice. *Pak. J. Bot.*, 46: 2015-2020.
39. Siddiqui, ZS. 2013: Effects of double stress on antioxidant enzyme activity in *Vigna radiata* (L.) Wilczek. *Acta Bot. Croat.* 72: 145-156.
40. Siddiqui, ZS and MA Khan. 2013: Some Physiological Attributes of Dimorphic Seeds of *Halopyrum Mucronatum* (L.) Stapf. *Pakistan Journal of Botany* 45:1975-1979.
41. Kwon, TR, SK Lee, SR Park, ZS Siddiqui, SJ Moon, SC Park, and MO Byun. 2013: *Atcbf1* Gene Enhances Salt Tolerance in Potato (*Solanum tuberosum* L.). *Plant Stress* 7: 34-38
42. Siddiqui, ZS and MA Khan. 2011: The role enzyme amylase in two germinating seed morphs of *Halophyrum mucronatum* (L.) Stapf in saline and non-saline environment. *Acta Physiologae Plantrum*. 33: 1185-1197.
43. Siddiqui, ZS and MA Khan. 2010: The role of seed coat phenolics on water uptake and early protein synthesis during germination of dimorphic seeds of *Halopyrum mucronatum* (L.) Stapf. *Pakistan Journal of Botany*. 42: 227-238.
44. Kwon, TR, ZS Siddiqui and PJC Harris. 2009: Effect of supplemental Ca^{++} on ion accumulation, transport and plant growth of salt sensitive *Brassica rapa* landraces. *Journal of Plant Nutrition*. 32: 644-667.
45. Siddiqui, ZS, MA Khan, K Beomgi, J-S Huang and TR Kwon. 2008: Physiological response of *Brassica napus* genotypes in combined stress. *Plant Stress*. 2: 78-83.

46. Kwon, TR, ZS Siddiqui and PJC Harris. 2008: Physiological variation of *Brassica* cultivars/landraces during germination and early seedling growth under salt stress. *Plant Stress*. 2: 103-109.
47. Siddiqui, ZS. 2007: Allelopathic potential of black pepper leachings on *Vigna mungo* (L.) Hepper. *Acta Physiologiae Plantarum*. 29:303-308.
48. Siddiqui, ZS. 2006: Biochemical responses of dimorphic seeds of *Arthrocnemum indicum* Willd. during germination, inhibition and alleviation under saline and non saline conditions. *Turkish Journal of Biology*. 30:185-193.
49. Siddiqui, ZS and S. Ahmed. 2006: Combined effect of pesticide on growth and nutritive composition soybean plants. *Pakistan Journal of Botany*. 38:721-733.
50. Siddiqui, ZS, Arif-Uz-Zaman and SS Shaukat. 2006: Alleviation of salinity induced dormancy by growth regulators in wheat seeds. *Turkish Journal of Botany*. 30:321-330.
51. Shaukat, SS and ZS Siddiqui. 2005: Effect of some organic nitrates on germination of *Cenchrus biflorus* Roxb Seeds. *International Journal of Biology & Biotechnology*. 2: 99-100.
52. Siddiqui ZS and Arif-Uz-Zaman. 2005: Effect of Capsicum leachate on germination and chlorophyll accumulation of *Vigna radiata* (L) Wilczek. *Pakistan Journal of Botany* 37: 941-947.
53. Habiba, R, ZS Siddiqui and Arif-Uz-Zaman. 2005: Changes in Chlorophyll, Protein and Carbohydrate contents of *Vigna radiata*, *Cicer aurietinum* and *Lens culinaris* after Irrigation with Sewage Waste Water. *International Journal of Biology and Biotechnology*. 2: 981-984.
54. Siddiqui, ZS, Habiba, R, and Arif-Uz-Zaman. 2005: Effects of chromium and lead on germination, accumulation and phenolic contents of *Gossypium hirsutum* (L.) and *Solanum melongena*. *International Journal of Biology and Biotechnology*. 2: 773-777.
55. Afzal, M., FS Mehdi, ZS Siddiqui and SS Shaukat. 2005: Use of *Rhizophora mucronata* Poir against the growth of some atmospheric fungal allerganes. *International Journal of Biology and Biotechnology*. 2: 941-945.
56. Siddiqui, ZS 2004: Effects of systemic fungicide (benlate) on germination, seedling growth, biomass and phenolic contents of *Vigna radiata* (L.) Wilczek and *Vigna mungo* (L.) Hepper. *International Journal of Biology and Biotechnology*. 1: 543-547.
57. Siddiqui ZS and Arif-uz Zaman. 2004: Effects of systemic fungicide (benlate) on germination, seedling growth, biomass and phenolic contents of two different varieties of *Zea mays*. *Pakistan Journal of Botany*. 36: 577-582.
58. Afzal, M., FS Mehdi, ZS Siddiqui and SS Shaukat. 2004: Periodicity in the Airborne Fungi of Karachi City. *International Journal of Biology and Biotechnology*. 1: 591-595.
59. Afzal, M., FS Mehdi and ZS Siddiqui. 2004: Effect of relative humidity and temperature on air borne fungal allergens of Karachi City. *Pakistan Journal of Biological Sciences*. 7:159-162.
60. Siddiqui, ZS and S. Ahmed. 2002: Effect of systemic fungicides on morphology, yield and nutritive composition of diseased and healthy plants of *Triticum aestivum* L. *Turkish Journal of Botany*. 26: 127-130.
61. Siddiqui, ZS and S. Ahmed. 2001: Effects of Methyl- Thiophenate (Systemic fungicide) on germination, seedling growth, biomass and phenolic content of resistant and susceptible varieties of *T. asetivum* L. *Pakistan Journal of Biological Sciences* 4: 1198-1200.

62. Siddiqui, Z.S and S. Khan 2001: Effects of systemic fungicide on absorption spectra, chlorophyll, protein and phenolic content of *V. radiata*. *Pakistan Journal of Biological Sciences*. 4: 812-814.
63. Neelofar, H., ZS Siddiqui and S. Ahmed. 2001: Effects of lead and copper on seed germination, accumulation and phenolic contents of *L. esculentum* and *S. oleracea*. *Pakistan Journal of Biological Sciences*. 4: 809-811.
64. Siddiqui, ZS and S. Ahmed. 2000: Effect of systemic fungicide on nutritive composition of diseased and healthy plant of *Triticum aestivum* L. *Pakistan Journal of Biological Sciences*. 3: 2148-2150.
65. Shaukat, SS., ZS Siddiqui and S. Aziz. 1999: Seed size variation and its effects on germination, growth and seedling survival in *Acacia nilotica* subspecies indica (Benth.) Brenan. *Pakistan Journal of Botany*. 31: 253-263.
66. Siddiqui, ZS, S. Ahmed and SS Shaukat. 1999: Effect of systemic fungicide (Topsin-M) and insecticide (Dimecron) on germination, seedling growth and phenolic content of *Pennisetum americanum*.L. *Pakistan Journal of Biological Sciences*. 2: 182-184.
67. Siddiqui, Z. S and S. Ahmed 1999: Effect of Dipterex insecticide on total carbohydrate, DNA, RNA and phenolic content of *Vigna radiata* (L.) Wilczek and *Vigna mungo* (L.) Hepper. *Pakistan Journal Botany*. 31: 93-96.
68. Shaukat, S. S., M. Mushtaq and Z. S. Siddiqui 1999: Effect of heavy metals on germination, seedling growth, biomass and phenolic content *P. aculeata* and *P. americanum*. *Pakistan Journal of Biological Sciences*. 2:1307-1313.
69. Siddiqui, ZS, S. Ahmed and Abid. 1999: Effect of systemic fungicides on germination, seedling growth, DNA, RNA and phenolic content of *B. compestris*. *Pakistan Journal of Biological Sciences*. 2: 1350-1351.
70. Siddiqui, ZS. 1997: Effect of systemic fungicides on total protein, carbohydrate and phenolic contents of *Solanum melongena* and *Avena sativa*. *Applied Entomology and Phytopathology*. 64:17-22.
71. Siddiqui, ZS, S. Ahmed and S. Gulzar. 1997: Effect of topsin-M (Methyl -thioophenate) Bayleton (Triademifon) on seedling growth, biomass, nodulation and phenolic content of *Sesbania sesban*. *Bangladesh Journal of Botany*. 26: 127-130.
72. Siddiqui, ZS and S. Ahmed. 1996: Effect of systemic fungicides on seed germination, seedling growth and phenolic content of *Vigna radiata*. *Pakistan Journal of Botany*. 28: 191-193.
73. Ahmed, S and ZS Siddiqui. 1995: Effect of topsin-M (Methyl-thiophenate) on chlorophyll, protein and phenolic content of *H.esculentus* and *Capsicum annuum*. *Pakistan Journal of Botany*. 26: 175-178.

EDITORIAL ROLE

EDITED MANUSCRIPTS (Frontier in Plant Sciences)

- Differential Metabolomic Responses of Kentucky Bluegrass Cultivars to Low Nitrogen Stress Xiaoyang Sun , Zhixin Guo ,Yiwei Jiang , Ligang Qin, Zhenjie Shi , Lili Dong, Liangbing Xiong , Runli Yuan , Wenjing Deng ,Hanfu Wu,Qingqing Liu, Fuchun Xie, Yajun Chen. **Frontiers in Plant Science Published on 28 Jan 2022.**
- Exogenous Melatonin Modulates Physiological Response to Nitrogen and Improves Yield in Nitrogen-Deficient Soybean (*Glycine max* L. Merr.) Huamei Wang, Chunyuan

Ren , Liang Cao , Qiang Zhao, Xijun Jin, Mengxue Wang, Mingcong Zhang, Gaobo Yu , Yuxian Zhang **Frontiers in Plant Science Published on 16 May 2022.**

- Melatonin-induced physiology and transcriptome changes in banana seedlings under salt stress conditions. Junya Wei, Jinhao Liang, Debing Liu, Yuewei Liu, Guoyin Liu, Shouxing Wei **Frontiers in Plant Science Published on 06 Sep 2022.**
- Iron and zinc micronutrients and soil inoculation of *Trichoderma harzianum* enhance wheat grain quality and yield. Iftikhar Ali, Ajab Khan, Ahmad Ali, Zahid Ullah, Dong-Qin Dai, Naveed Khan, Asif Khan, Abdel Rahman Al-Tawaha, Hassan Sher. **Frontiers in Plant Science Published on 06 Sep 2022.**
- Prolonged drought regulates the silage quality of maize (*Zea mays* L.): Alterations in fermentation microecology. Huamei Wang, Chunyuan Ren, Liang Cao Qiang, Zhao Xijun, Jin_Mengxue, Wang Mingcong, Zhang Gaobo Yu, Yuxian Zhang. **Frontiers in Plant Science Published in Dec 2022.**
- A microencapsulation approach to design microbial seed coatings to boost wheat seed germination and seedling growth under salt stress. Min Gong, Jiuxing He , Meng Kong, Qiuyan Huo , Yawen Jiang , Jiqing Song , Wei Han and Guohua L **Front. Plant Sci., 21 November 2023**

BOOK CHAPTERS

- 1) S. Gul, ZS.Siddiqui, A. Noman and F. Rasheed. 2018. Chapter 3. Tree Anatomy and Physiology. In A textbook of Applied Forestry.
- 2) Zamin Shaheed Siddiqui*, Muhammad Umar*, Taek-Ryoun Kwon, Soo Chul Park 2019: Chapter “Phenotyping through Infra-Red Thermography iPan stress environment”. In SUBKHA. Springer Nature

LIST OF GRANT / CONFERENCES/ WORKSHOP

1. Biochemical Responses of Halophyte during germination. **(Dean Faculty Science UOK)**
 2. Effect of systemic insecticide on growth, nutritive composition, and yield of *Abelmoschus esculentus* L. **(Dean Faculty Science UOK)**
 3. The role of seed coat phenolics in diverse germination responses of *Halopyrum mucronatum* (L.) Stapf. during germination. **(Dean Faculty Science UOK)**
 4. Physiological responses of *Branssica napus* genotypes under combined salt and drought stress. **(Dean Faculty Science UOK)**
 5. Phenotyping of some oil seed crops using IR thermography. **National Research Program for Universities, HEC, Pakistan. HEC PROJECT ID # 6586 dated 2016-17**
-

INTERNATIONAL RESEARCH PROJECTS

- 1) Physiological characterization and field assessment of Korean rice in Pakistan. (In Progress). This work is supported by a grant from the NEXT- GENERATION BIO-GREEN 21 Program No. **PJ009008** and **PJ009505**, **Rural Development Administration, Suwon, Republic of Korea (Completed 2013-16)**
-

THESIS STUDENTS

Ph.D.

1. **Roomana Yasmeen** 2018 “Physiological responses of crop plants against *Trichoderma harzianum* in saline environment”. **(Awarded)**
2. **Muhammad Umar** 2018 “Physiological assessment of sunflower cultivar against abiotic stress”. **(Awarded)**
3. **Amrina Habib** 2019 “Study on the effect of *Pythium aphanidermatum* on the physiology of *Luffa cylindrica* and its management by microbial antagonists” **(Awarded)**
4. **Ayesha Tabassum** 2021 “Assessment of Phytoremediation potential of *Daucus Carota* L. (Carrot) for Zn and Cu toxicity through physiological markers” **(Awarded)**
5. **Shumaila Iqbal** Impact of Silica and titanium nanoparticles on Growth and Physiology of Crop plants under stress conditions
6. **Nazia Emad** Mitigating effects of nanoparticles in the physiology of crop plants under saline habitat.
7. **Fiza Ali**: Effect of some plant growth promoting microorganism on photochemical efficiencies energy fluxes nitrogen assimilation of sunflower genotypes in salt stress environment.

M. Phil

1. **Huda Shahid** 2016: “Physiological screening and field assessment of some rice genotypes under optimal and suboptimal environments **(Awarded)**
 2. **Fiza Ali** 2018 “Photochemical Screening of Some leguminous varieties against drought stress environment using plant growth promoting bacteria **(Awarded)**
 3. **Basit Ali** 2021 “Physiological responses of halophytic and non-halophytic plants in abiotic stress environment **(Awarded)**
 4. **Mahnoor** 2022-23 Photochemical screening of crop plant using PGPM in heavy metal contaminated soil **(Awarded)**
 5. **Hira Naz** 2022-23 Physiological role of Endophytes in crop plants against salinity **(Awarded)**
 6. **Naeema Asim** 2023 Systemic efficacy of systemic fungicides against plant diseases and its control management.
 7. **Yusra Rehman** 2023 Comparative Physiological Response of Plant Growth Promoting Microorganism (PGPM) and Fungicide under Fungal Stressed Crop Plant **(Awarded)**
 8. **Yumna Rao** 2023 Effects of algal biochar on growth and physiology of rice under salt stress **(Awarded)**
 9. **Ayesha ftikhar** 2023 Nonfertilizer and its application on crop plant under saline habitat **(Awarded)**
 10. **Hafiza Hamna Ansari** 2023: Combined application of nanoparticles and plant growth promoting microbes to enhance salt tolerance in Zea Mays (Maize)
-

11. **Ms Umm Ey Farwa Hassan 2023:** *Effect of compost on growth and eco physiological performance of Maize plants in salinity (Awarded)*
 12. **Hafiza Khadija Irshad 2023:** *Exogenous application of elicitors on crop plants under different moisture stress. (Submitted)*
-

EDITORIAL CONTRIBUTION

a) Journals Reviewer

- 1) *Acta Physiologia Plantarum SPRINGER*
- 2) *Plant Biology BMC*
- 3) *African Journal of Agricultural Science*
- 4) *Pakistan Journal of Botany*
- 5) *Horticulture Sci*
- 6) *Environmental Science and Pollution Research SPRINGER*
- 7) *Photosynthetica*
- 8) *Journal of Plant Growth Regulation SPRINGER*
- 9) *Scientific Reports SPRINGER NATURE*
- 10) *Letter in Applied Microbiology JOHNS AND WILEY*

b) Associate Editor

- 1) *Frontier in Plant Sciences*
-

Thesis Completed

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|-------------------|-------------------|
| 1) MSc: | Total 50 Students |
| 2) M Phil: | Total 08 Students |
| 3) PhD : | Total 04 Students |

REFERENCE

Dr S. M Shahid

Senior Post Graduate Lecturer and Research Supervisor
School of Health Science, Eastern Institute of Technology
(EIT)-Te Pukenga , 238 Queen Street, Auckland
Email: sshahid@eit.ac.nz
Tel: 0064220260092

Dr. Jung Il Cho 조정일

National Center for GM Crop
Rural Development Administration Republic of Korea
Email: jungilcho@korea.kr

Dr.Taek-Ryoun Kwon (Director General),

National Center for GM Crop, National Institute of Agricultural Biotechnology,
Rural Development Administration, South Korea
Email: trkwon@gmail.com